

Judge Questions and Answers for UPEX 2026 Poster

Coral Reef Health Assessment | concise defense notes

Is the original Q&A enough? Partly. It covers the basic story, but a judge may ask harder questions about validation, overfitting, confidence calibration, dataset bias, Grad-CAM reliability, and ReefGuide limitations. This version adds those missing defense points.

Fast Defense Strategy

Defense Focus	Best Message
Start with impact	Faster coral health screening for conservation support.
Defend the method	EfficientNetB0, stratified split, augmentation, class balancing, ensemble, SWA, TTA.
Be honest on limits	Strong held-out performance, but external field validation is still needed.
Build trust	Use confidence calibration, uncertainty flag, and Grad-CAM explanation.
Avoid overclaiming	Position the platform as decision support, not a full replacement for experts.

1. Project Overview and Value

Q&A	Can you briefly explain your project and the problem it solves? - Problem: coral monitoring is slow, costly, and depends heavily on expert manual inspection. - Solution: a web platform that classifies coral images into Healthy, Bleached, and Dead. - Key value: combines CNN prediction with Grad-CAM explanation, so users can see why the model made a decision. - Practical use: supports faster reef-health screening for researchers, divers, and conservation teams.
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Q&A	What is the main novelty of your project? - Not only classification; the system also provides explainability through Grad-CAM. - It includes an end-to-end pipeline: data preparation, model training, evaluation, deployment, and ReefGuide assistant. - The web interface makes the model usable by non-technical users.
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Q&A	Why should this project matter to environmental agencies? - It can reduce the time needed to screen coral images. - It can help prioritize reef areas that need expert attention. - It provides visual evidence, confidence score, and chatbot explanation to support decision-making.
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2. Dataset and Validation

Q&A	How did you prevent data leakage? - Images were split into train, validation, and test sets before final evaluation. - The test set was kept unseen during training. - The split was saved in split_info_v3.json so repeated training uses the same split. - Stratified splitting keeps class ratios consistent across sets.
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Q&A	Your dataset is imbalanced. How did you handle that? - Used stratified splitting to preserve class distribution. - Used hard-example oversampling so difficult samples appear more often during training. - Used class weighting so mistakes on minority or critical classes receive higher penalty. - Dead class penalty was increased by 1.3x because missing Dead coral is high risk.
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Q&A	Is 98.11% accuracy enough to prove real-world readiness? - It proves strong performance on the held-out test set, but not full real-world readiness. - Real-world deployment still needs larger external validation from different reef locations, cameras, and water conditions. - We reduce risk using TTA, confidence calibration, Grad-CAM, and an uncertainty flag. - For field use, this should support expert review, not replace marine experts completely.
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Q&A	What is the biggest risk in your dataset? - Dataset bias: images may not represent all reef species, water conditions, camera types, or geographic regions. - Visual overlap: severely bleached and newly dead coral can look similar. - Future work: expand the dataset with more locations, more disease classes, and expert-confirmed labels.
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3. Model Design and Training

Q&A	Why did you choose EfficientNetB0 instead of ResNet, VGG, or a bigger model? - EfficientNetB0 balances accuracy and efficiency well. - It is lightweight, so inference is faster for a web application. - It uses fewer parameters than heavier architectures, reducing training and memory cost. - Transfer learning from ImageNet helps because the dataset is limited.
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Q&A	Why did you fine-tune only the top 100 layers? - Early layers learn general visual features such as edges and textures. - Top layers learn task-specific coral features. - Fine-tuning only upper layers reduces overfitting and training cost. - It adapts the model without destroying useful pretrained knowledge.
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Q&A	Why use augmentation and mixup? - Coral images vary by angle, distance, lighting, turbidity, and water color. - Augmentation simulates these changes during training. - Mixup softens decision boundaries and reduces sensitivity to label noise. - Together, they make the model less likely to memorize the training images.
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Q&A	How did you reduce overfitting? - Used data augmentation and mixup. - Used dropout 0.4 and L2 regularization 0.0002. - Used transfer learning instead of training from scratch. - Used SWA and ensemble prediction for better generalization.
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Q&A	Why train five models with different seeds? - A single model can be affected by random initialization and training variation. - A 5-seed ensemble reduces variance. - Averaging model outputs gives more stable predictions and confidence scores.
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Q&A	What is SWA, and why did you apply it? - SWA means Stochastic Weight Averaging. - It averages model weights from the final training epochs. - This moves the model toward a flatter minimum. - Flatter minima usually generalize better to unseen data.
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4. Evaluation, Confidence, and Explainability

Q&A	What is Test-Time Augmentation (TTA), and why use it? - TTA creates several versions of the same test image. - This project uses original and flipped views at 224 and 256 center-crop scale. - Each image receives predictions from 5 models and 4 TTA views, giving 20 predictions. - Averaging these predictions makes the final result more stable.
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Q&A	Why use temperature scaling? - Neural networks can be overconfident. - Temperature scaling calibrates the probability values. - The goal is to make confidence closer to real expected correctness. - In this project, the temperature parameter is $T = 0.441$.
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Q&A	How does Grad-CAM solve the black-box problem? - Grad-CAM produces a heatmap showing image regions that influenced the prediction. - It lets users check whether the model focuses on coral tissue instead of background noise. - Averaged Grad-CAM across the ensemble improves stability. - Eigen-smooth denoising makes the heatmap cleaner.
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Q&A	Can Grad-CAM prove the model is always correct? - No. Grad-CAM explains model attention; it does not guarantee correctness. - It is used as supporting evidence for trust and review. - If Grad-CAM focuses on irrelevant background, users should treat the prediction carefully.
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Q&A	Which classes are hardest to distinguish? - Bleached versus Dead can be difficult because both may appear pale or low-color. - Newly dead coral and severely bleached coral can be visually subtle even for humans. - This is why oversampling, class weighting, ensemble prediction, and Grad-CAM are important.
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5. Deployment and ReefGuide

Q&A	How does the deployed system work? - The user uploads a coral image through the React frontend. - The Flask backend preprocesses the image and calls /api/predict. - The system returns class prediction, confidence, probabilities, original image, Grad-CAM overlay, and uncertainty flag. - ReefGuide can explain the result in user-friendly language.
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Q&A	Why include a chatbot? - Non-expert users may not understand probabilities and Grad-CAM heatmaps. - ReefGuide translates the model output into short ecological explanations. - It uses the latest prediction context and chat history to answer relevant questions.
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Q&A	What are the limitations of ReefGuide? - It depends on the model output and provided context. - It should not invent scientific conclusions outside the prediction evidence. - It should be treated as an explanation assistant, not a replacement for expert field assessment. - API key and prompt safety must be managed carefully in deployment.
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Q&A	What happens if the model confidence is low? - The system marks uncertainty when confidence is below the threshold. - Low-confidence cases should be reviewed by a human expert. - This is safer than forcing every image into a high-confidence decision.
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6. Limitations and Future Work

Q&A	What are your main current limitations? - The system uses static 2D images. - Performance depends on dataset coverage and label quality. - It currently classifies only three broad classes: Healthy, Bleached, and Dead. - External validation is still needed for different reef regions and camera conditions.
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Q&A	What would you improve next? - Add more coral health classes and disease categories. - Collect larger expert-labeled datasets from multiple reef locations. - Support real-time video from underwater drones. - Add geolocation and time-series tracking for reef monitoring.
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Q&A	If the judge challenges your result, what is the safest answer? - Say the result is strong for the held-out test set, but field deployment needs external validation. - Emphasize that the platform is a decision-support tool. - Mention safeguards: confidence calibration, uncertainty flag, Grad-CAM explanation, and human expert review.
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Questions You Must Prepare Numbers For

- Final test accuracy and class-level precision, recall, and F1-score.
- Confusion matrix: especially Bleached vs Dead mistakes.
- Dataset split counts for Healthy, Bleached, and Dead.
- Demo example: one confident case and one uncertain case.